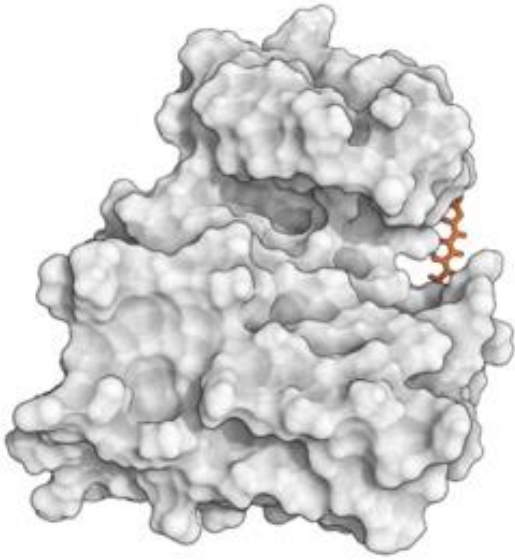
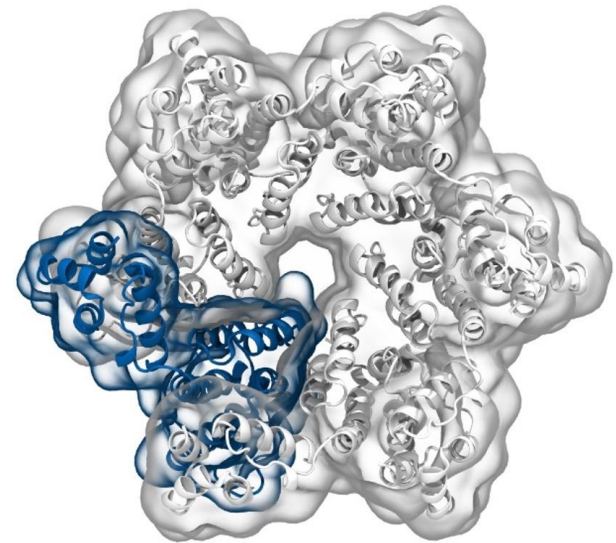


Simulation of Biomolecules



Setting up a protein simulation



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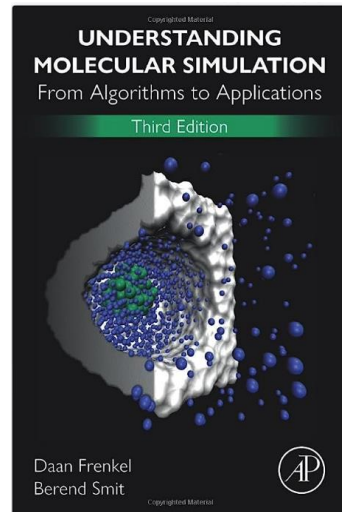
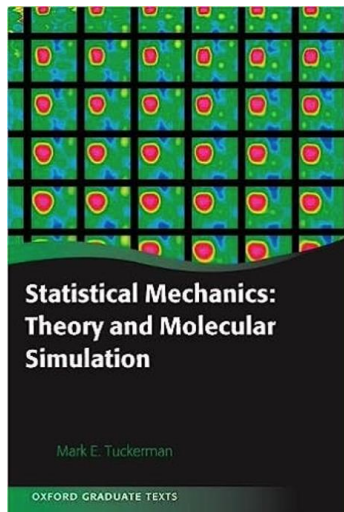
Useful resources to learn running simulations

Best Practices for Foundations in Molecular Simulations [Article v1.0]

Efrem Braun¹, Justin Gilmer², Heather B. Mayes³, David L. Mobley⁴, Jacob I. Monroe⁵, Samarjeet Prasad⁶, Daniel M. Zuckerman⁷

A suite of tutorials for the BioSimSpace framework for interoperable biomolecular simulation [Article v1.0]

Lester O. Hedges^{1,2*}, Sofia Bariami^{3†}, Matthew Burman², Finlay Clark³, Benjamin P. Cossins⁴, Adele Hardie³, Anna M. Herz³, Dominykas Lukauskis⁵, Antonia S.J.S. Mey³, Julien Michel^{2,3*}, Jenke Scheen^{3‡}, Miroslav Suruzhon⁴, Christopher J. Woods¹, Zhiyi Wu⁴



From Proteins to Perturbed Hamiltonians: A Suite of Tutorials for the GROMACS-2018 Molecular Simulation Package [Article v1.0]

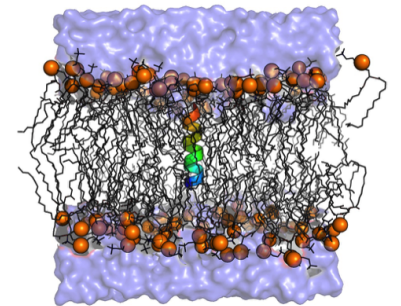
Justin A. Lemkul

Department of Biochemistry, Virginia Polytechnic Institute and State University

<https://orcid.org/0000-0001-6661-8653>

DOI: <https://doi.org/10.33011/livecoms.1.1.5068>

Keywords: tutorials, gromacs, molecular dynamics simulation, computational chemistry



 PDF

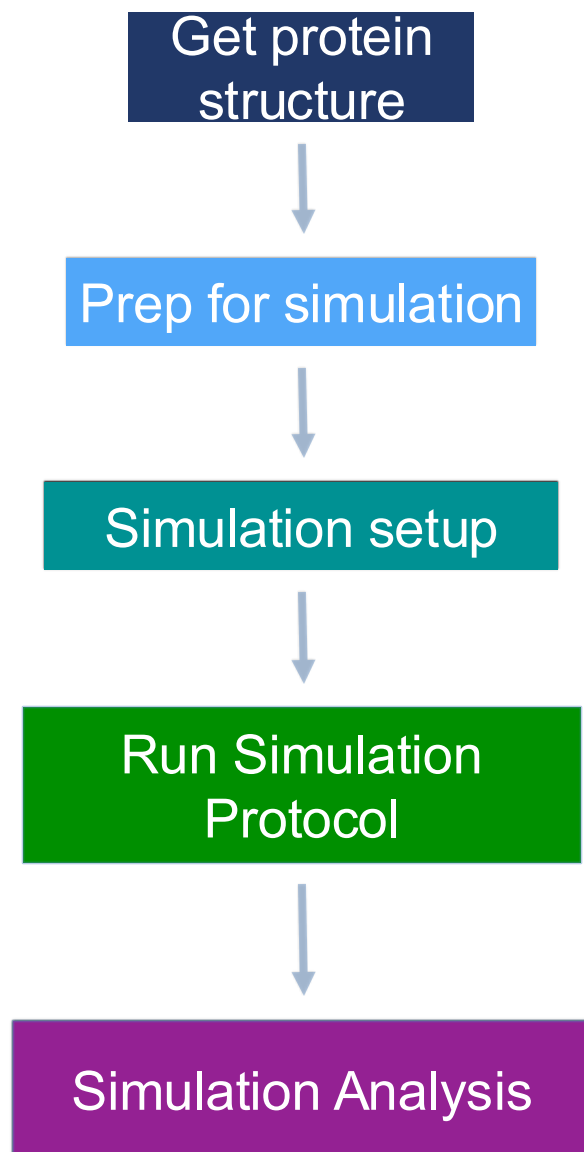
 ARTICLE CODE REPOSITORY

GROMACS: tutorials.gromacs.org

Amber: ambermd.org/tutorials

OpenMM: <https://openmm.github.io/openmm-cookbook/dev/index.html>

A typical workflow for molecular dynamics



Getting your protein structure

AlphaFold Protein Structure Database

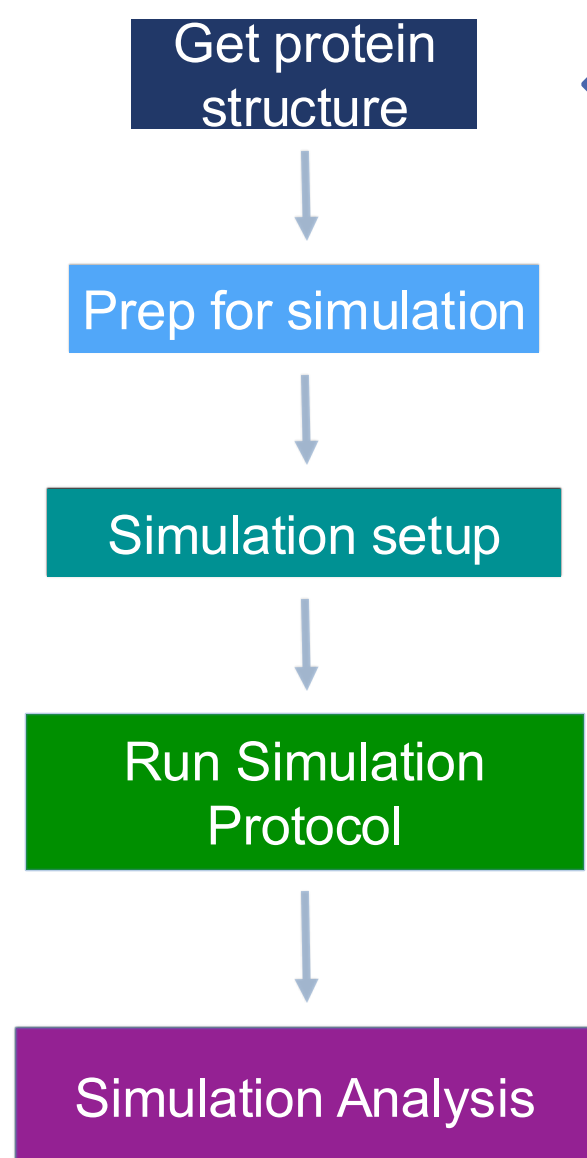


Getting ligands/co-factors

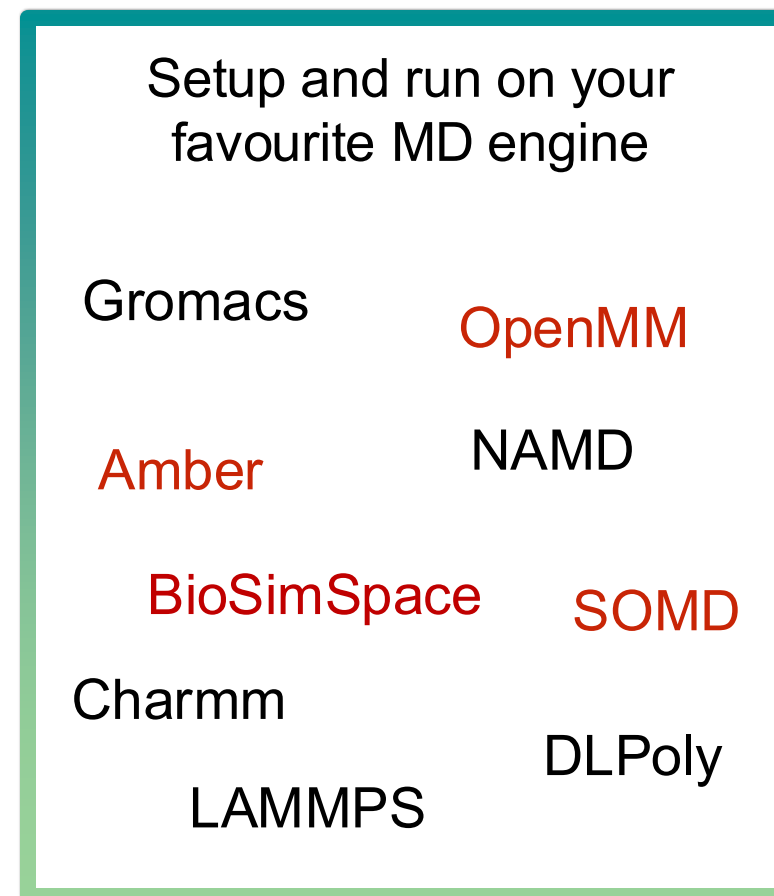
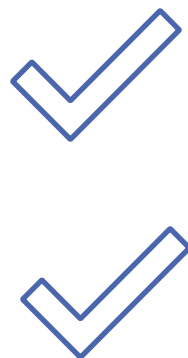
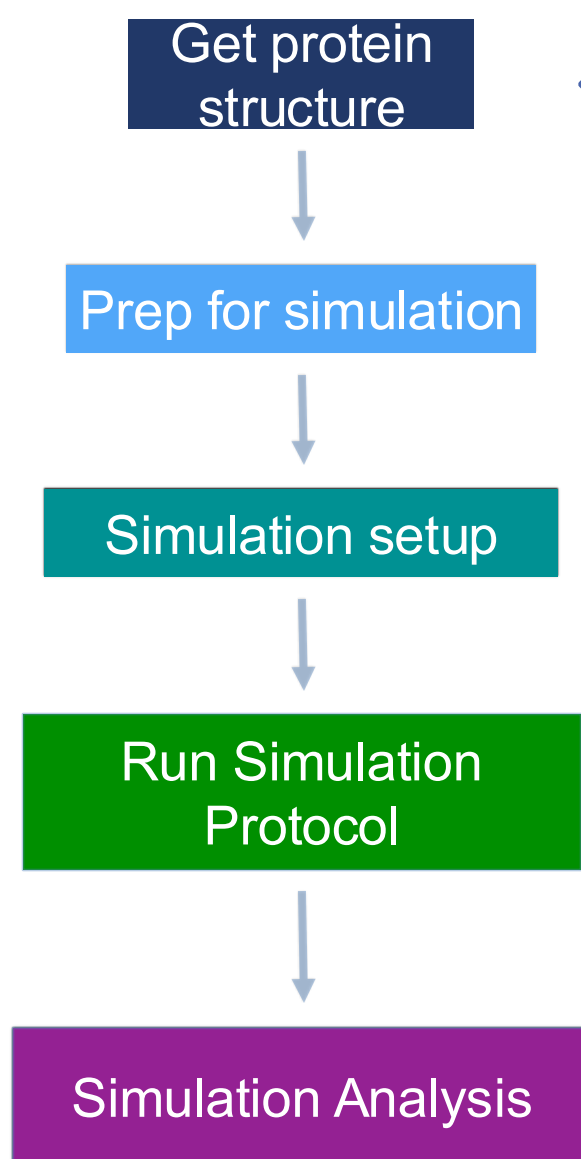
ZINC20



A typical workflow for molecular dynamics

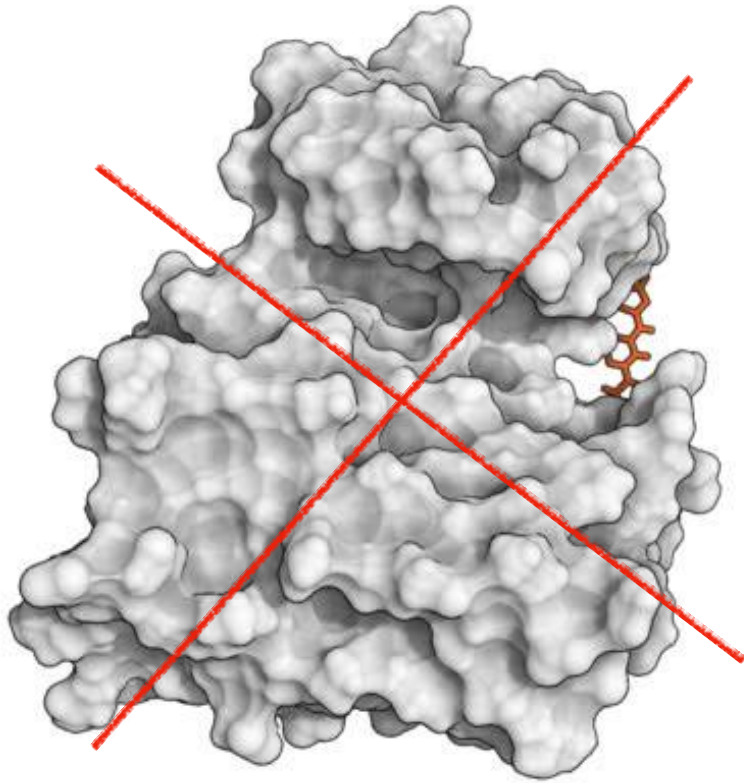


A typical workflow for molecular dynamics



*mostly C++
command line* *Python API*

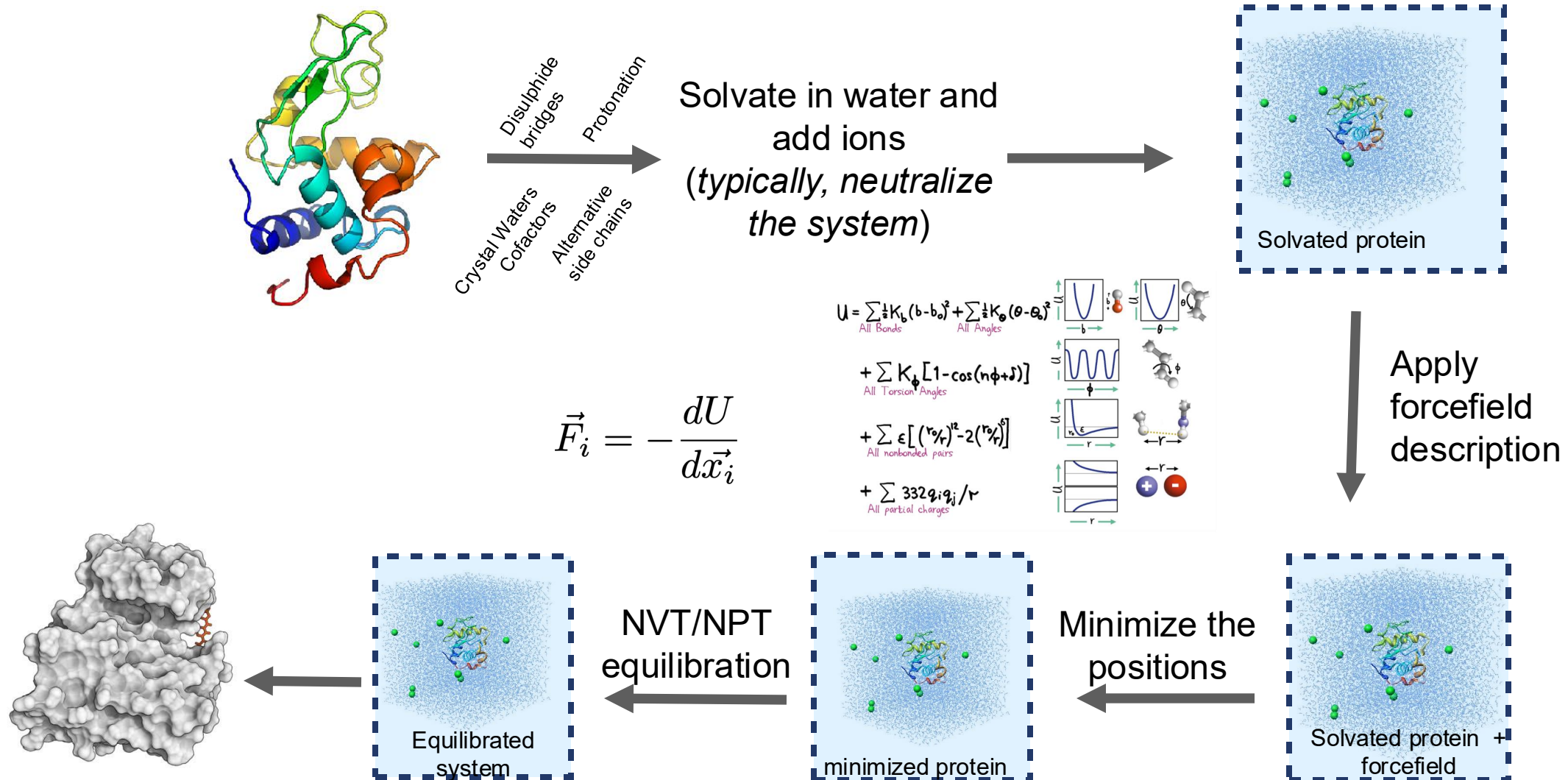
Disclaimer!



Running biomolecular MD can take days on specialised hardware.

Today we will *not* run any of them, and instead will focus on fundamental principles using small molecules.

Molecular dynamics require multiple steps for the setup of simulations

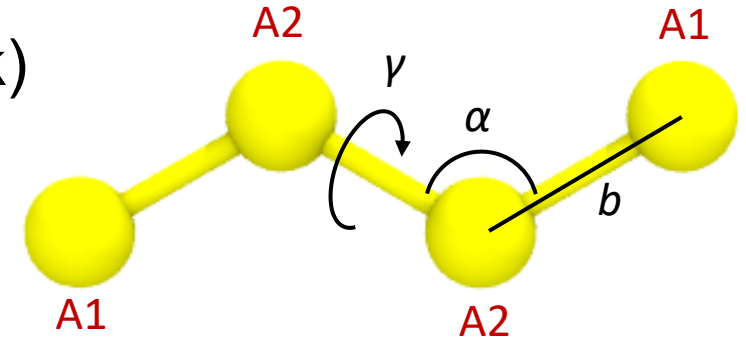


Production simulation

Evaluating interatomic interactions

MD simulation require defining a system *topology*. This includes information on:

- *Connectivity* between atoms (covalent bond network)
- *Atom types* (depend on chemical environment, e.g., sp² or sp³ Carbon)



$$\vec{F}_i = -\frac{dU}{d\vec{x}_i}$$

$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_0)^2 + \sum_{\text{All Angles}} \frac{1}{2} K_\theta (\theta - \theta_0)^2$$

$$+ \sum_{\text{All Torsion Angles}} K_\phi [1 - \cos(n\phi + \delta)]$$

$$+ \sum_{\text{All nonbonded pairs}} \epsilon \left[\left(\frac{r_0}{r} \right)^{12} - 2 \left(\frac{r_0}{r} \right)^6 \right]$$

$$+ \sum_{\text{All partial charges}} \frac{332 q_i q_j}{r}$$

The diagrams illustrate the functional forms of interatomic interactions:

- Bond stretching:** A parabolic potential energy curve U versus bond length b , with a minimum at b_0 .
- Angle bending:** A parabolic potential energy curve U versus bond angle θ , with a minimum at θ_0 .
- Torsion:** A periodic potential energy curve U versus torsion angle ϕ , showing maxima and minima.
- Lennard-Jones potential:** A potential energy curve U versus distance r , showing a minimum at r_0 .
- Electrostatic interactions:** A potential energy curve U versus distance r for partial charges q_i and q_j , showing a minimum at r_0 .

The functional form of interatomic interactions and their strengths is defined by a *force field*.

Many force fields are available

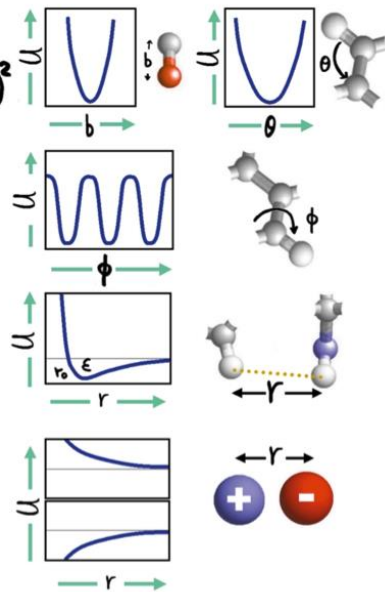
Biomolecular force fields

$$\vec{F}_i = - \frac{dU}{d\vec{x}_i}$$

$$U = \sum_{\text{All Bonds}} \frac{1}{2} K_b (b - b_0)^2 + \sum_{\text{All Angles}} \frac{1}{2} K_\theta (\theta - \theta_0)^2$$

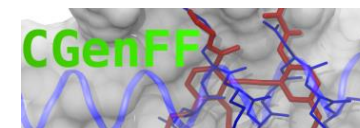
$$+ \sum_{\text{All Torsion Angles}} K_\phi [1 - \cos(n\phi + \delta)]$$

$$+ \sum_{\text{All nonbonded pairs}} \epsilon \left[\left(\frac{r_0}{r} \right)^{12} - 2 \left(\frac{r_0}{r} \right)^6 \right]$$

$$+ \sum_{\text{All partial charges}} \frac{332 q_i q_j}{r}$$


- **Amber** (incl. Glycam params for glycans)
- **CHARMM** (incl. POPC, POPE, DPPC lipids)
- **OPLS**
- **GROMOS**
- ...

Small molecule force fields



$$E_{\text{pair}} = \sum_{\text{bonds}} K_r (r - r_{\text{eq}})^2 + \sum_{\text{angles}} K_\theta (\theta - \theta_{\text{eq}})^2 +$$

$$\sum_{\text{dihedrals}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[\frac{A_{ij}}{R_{ij}^{12}} - \frac{B_{ij}}{R_{ij}^6} + \frac{q_i q_j}{\epsilon R_{ij}} \right]$$

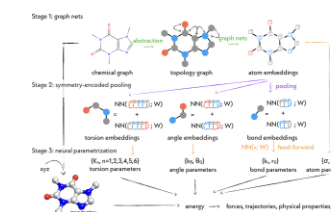
GAFF

Machine learned force fields



Espaloma

Ani-2x



SchNet

PhysNet

BandNN

MACE-OFF

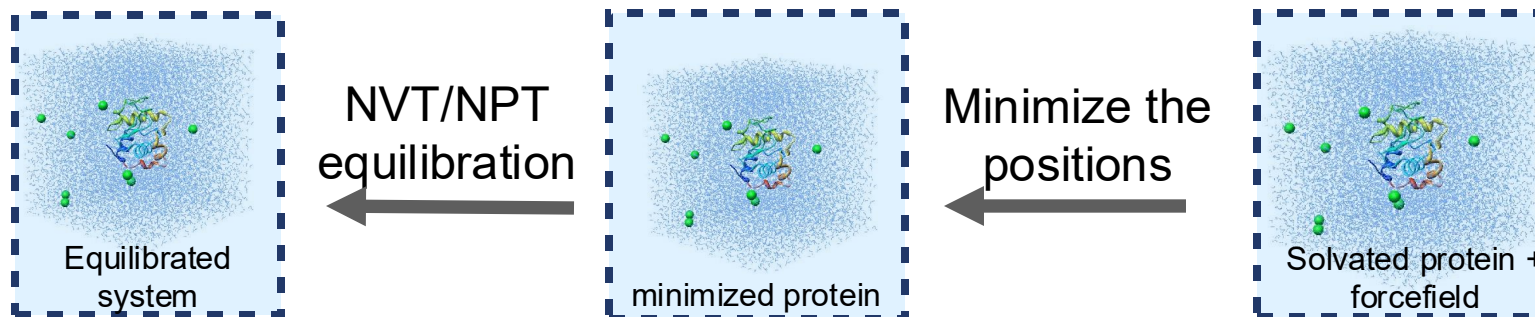
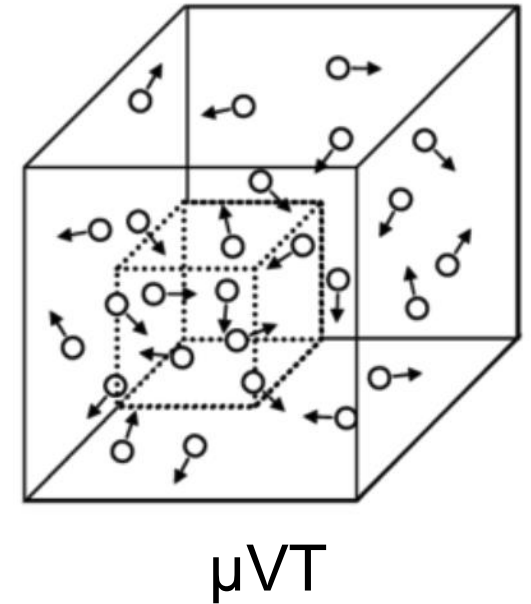
There is no “best force field”!

Choosing your thermodynamic ensemble

Simulations replicate a specific *thermodynamic ensemble* (typically NVT or NPT), or even grand canonical (μ VT)

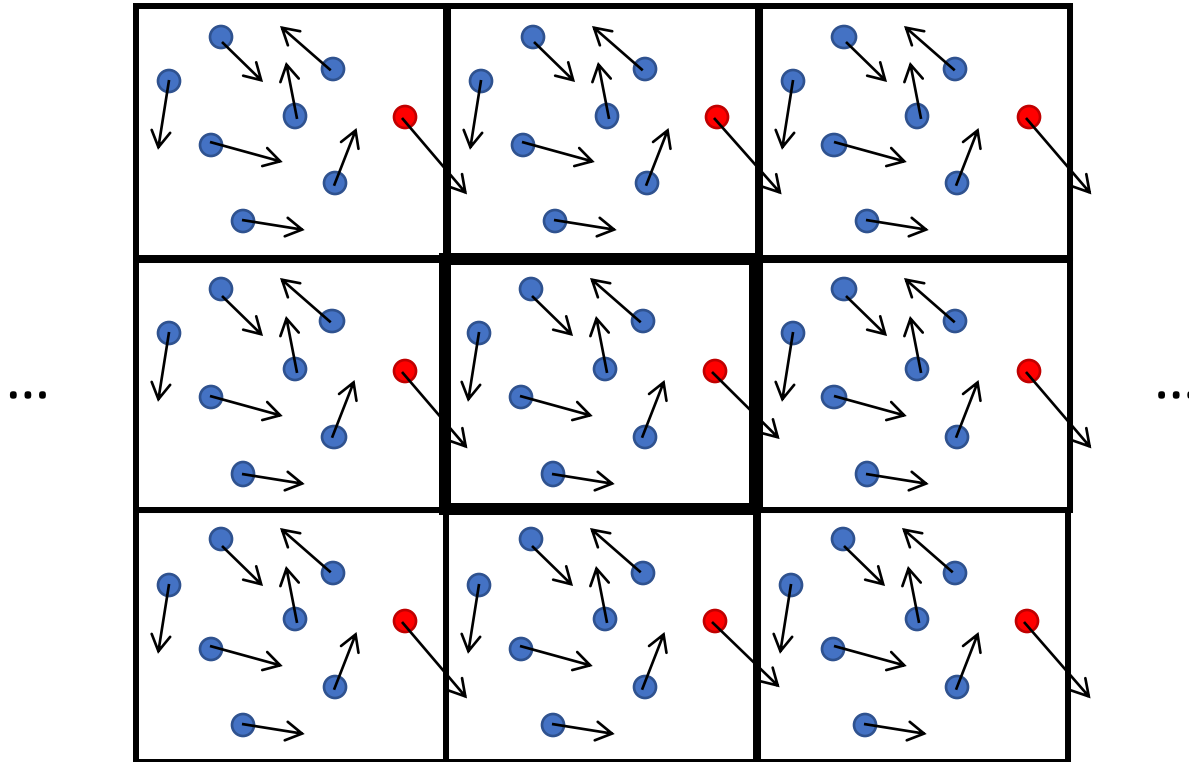
You will have different options to include *thermostats* (scaling atom velocities) and *barostats* (scaling positions) in your calculations:

- Nose-Hoover
- Berendsen
- Parrinello-Rahman
- Langevin piston
- ...

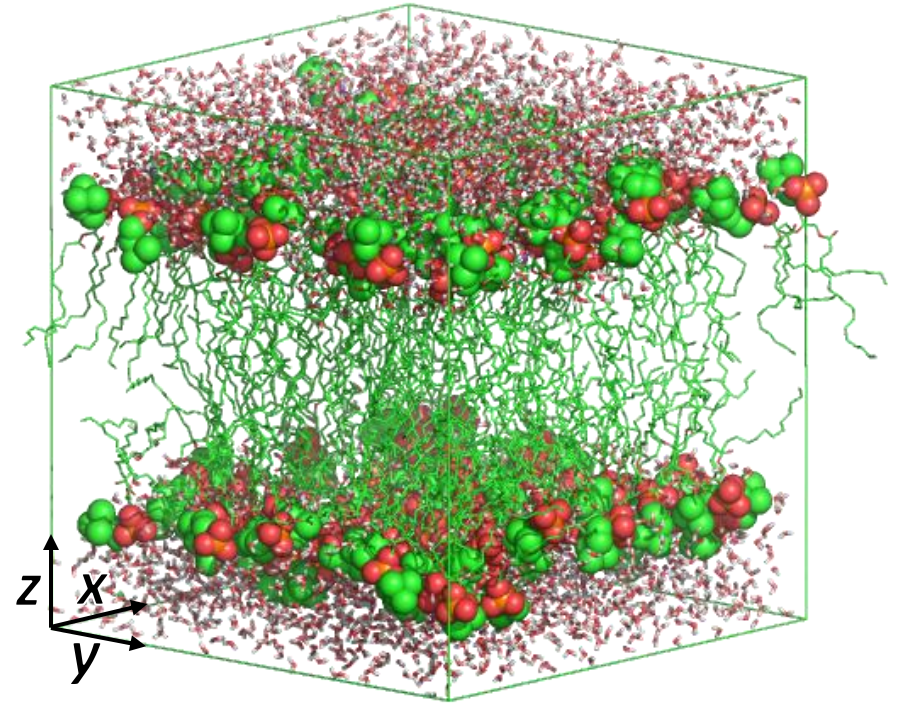


Periodic boundary conditions (PBC) and pressure coupling

Useful to reduce finite-size effect and simulate bulk

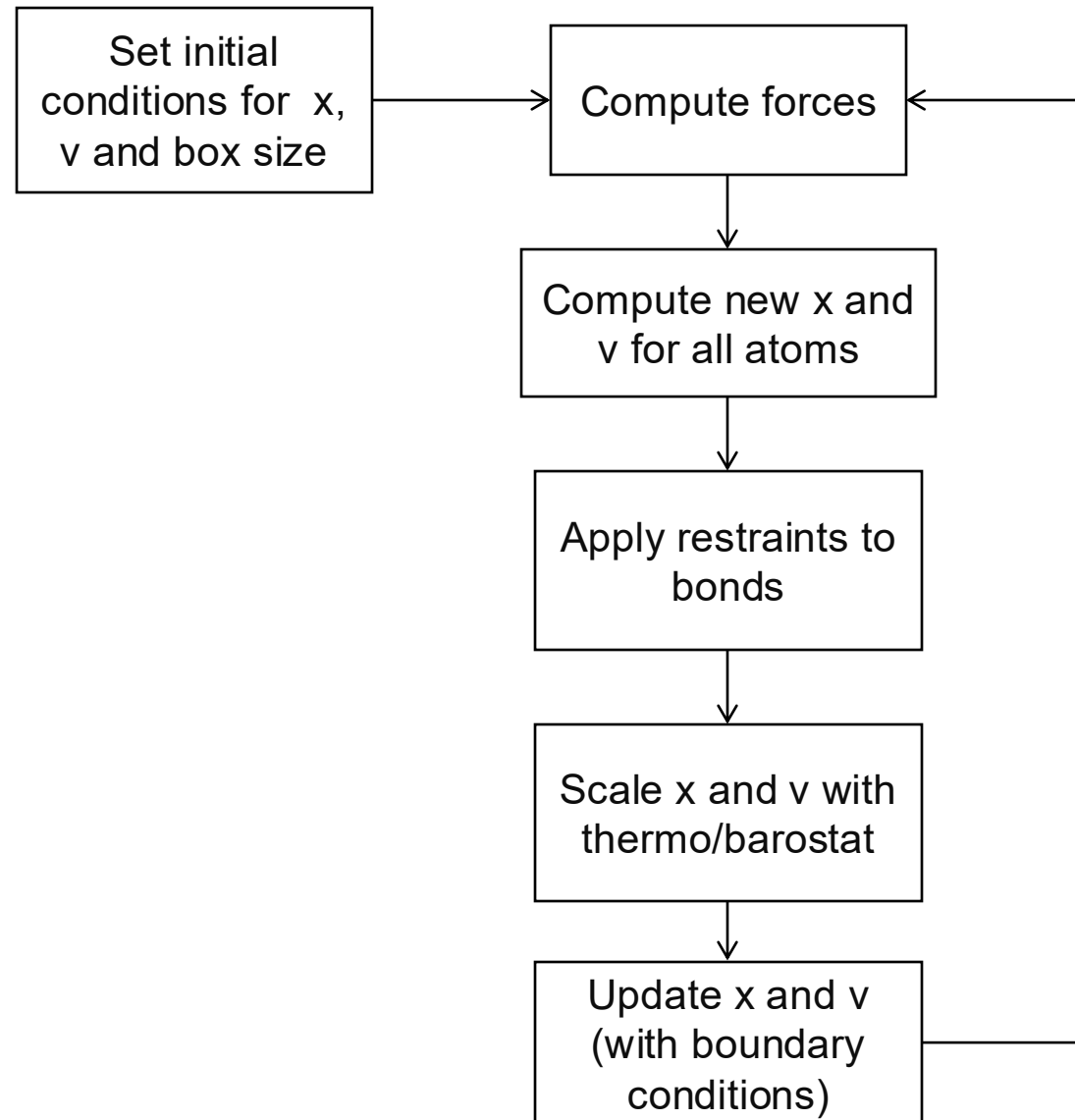


Typically, PBC applied in x, y and z direction



For membrane systems, use semi-isotropic pressure coupling ($x, y \neq z$, lipids compressibility is direction-dependent)

A Molecular Dynamics timestep



Sampling timescales for protein systems

The steepest gradient determines the smallest timestep.

Timestep size is imposed by the fastest phenomenon we want to observe.

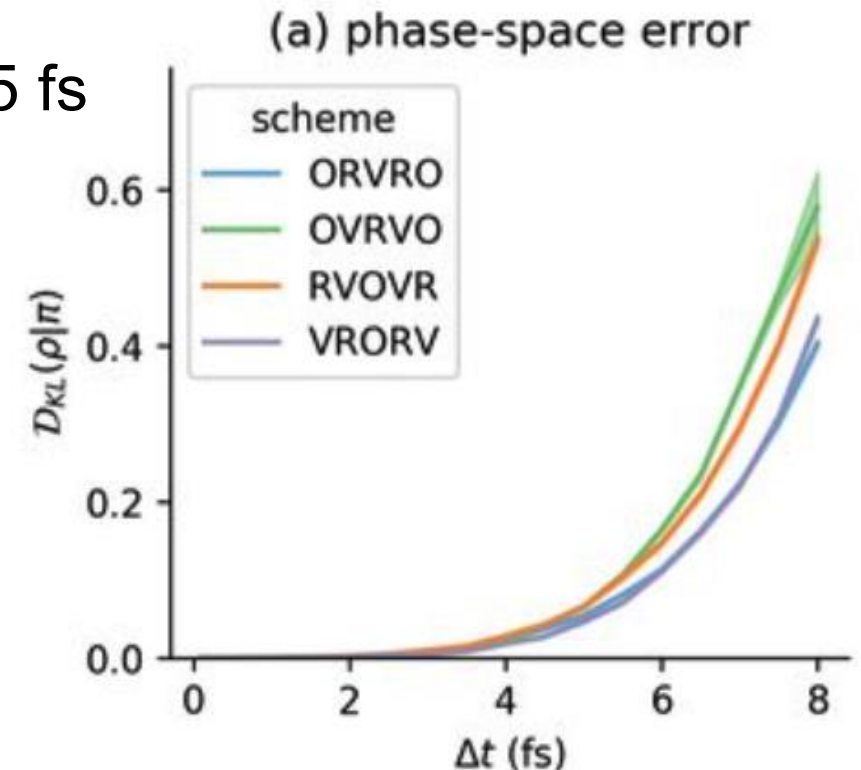
In atomistic simulations:

- Covalent bond hydrogen-heavy atom (10^{14} Hz): 0.5 fs
- Covalent bond heavy atom-heavy atom: 1 fs
- Angles fluctuations: 2 fs

Restraining covalent bond distances allows to use 1-2 fs timesteps (restraining methods: SHAKE, RATTLE, LINCS,...)

Hydrogen Mass repartitioning: 4 fs

Other integrators (e.g., Langevin): 4 fs - 6 fs.



“equilibration” and “convergence”: what do they mean?

Equilibration phase:
is the system in a
“relaxed” state?

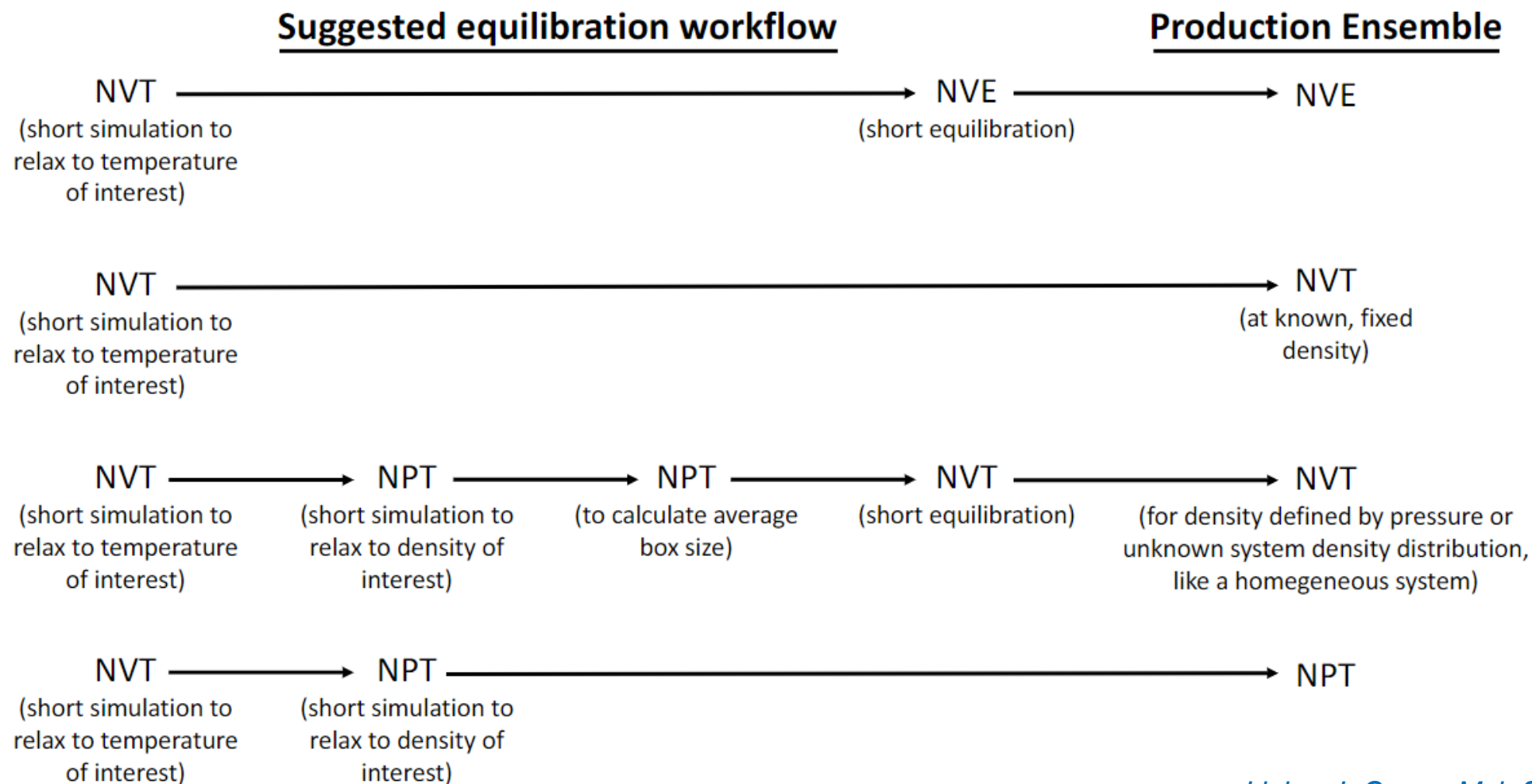
Production phase: do
we have good sampling
and convergence?



Thinking about the problem holistically: an integrated framework for the analysis of equilibration, sampling, and convergence.

Example equilibration protocols

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)



An *example* simulation protocol

YOU WANT: constant volume, pressure, and temperature, healthy Ramachandran plot, no exotic chemistry, bulk water (if used)

Equilibration:

1. Minimize energy, 1000 steepest descent
2. Heat system from 0 to 300 K in 500 ps, NPT, Berendsen barostat 1 atm. α -carbon restrained with 10 kcal/mol harmonic potential. 2 fs timestep, LINCS all bonds
3. 1 ns NVT equilibration with Langevin dynamics, no atom constrained.

Production:

4. 1 μ s NPT, Nose-Hoover barostat, PME for electrostatics

DETERMINE HANDLING OF CUTOFFS

- ☐ As a general rule, electrostatics are long-range enough that either the cutoff needs to be larger than the system size (for finite systems) or periodicity is needed along with full treatment of long-range electrostatics (Section 3.4)
- ☐ Nonpolar interactions can often be safely treated with cutoffs of 1-1.5 nm as long as the system size is at least twice that, but long-range dispersion corrections may be needed (Section 4.1)

CHOOSE APPROPRIATE SETTINGS FOR THE DESIRED ENSEMBLE

- ☐ Pick a thermostat that gives the correct distribution of temperatures, not just the correct average temperature; if you have a small system or a system with weakly interacting component choose one which works well even in the small-system limit.
- ☐ Pick a barostat that gives the correct distribution of pressures
- ☐ Consider the known shortcomings and limitations of certain integrators and thermostats/barostats and whether your choices will impact the properties you are calculating

CHOOSE AN APPROPRIATE TIMESTEP FOR STABILITY AND AVOIDING ENERGY DRIFT

- ☐ Determine the highest-frequency motion in the system (typically bond vibrations unless bond lengths are constrained)
- ☐ As a first guess, set the timestep to approximately one tenth of the highest-frequency motion's characteristic period
- ☐ Test this choice by running a simulation in the microcanonical ensemble, and ensure that energy is conserved

Which MD engine should I use?

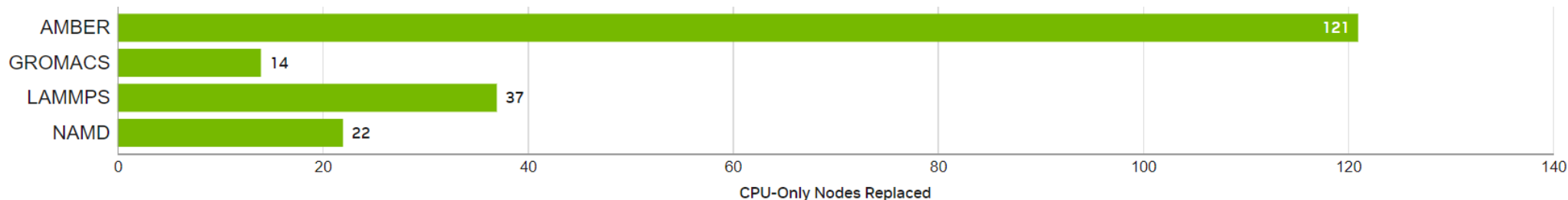
Consider:

- Support for force field of choice
 - Enables running desired simulation protocol
 - Performance for available hardware
 - Ease of use
- depends on number of atoms, hardware, simulation protocol, MD engine*

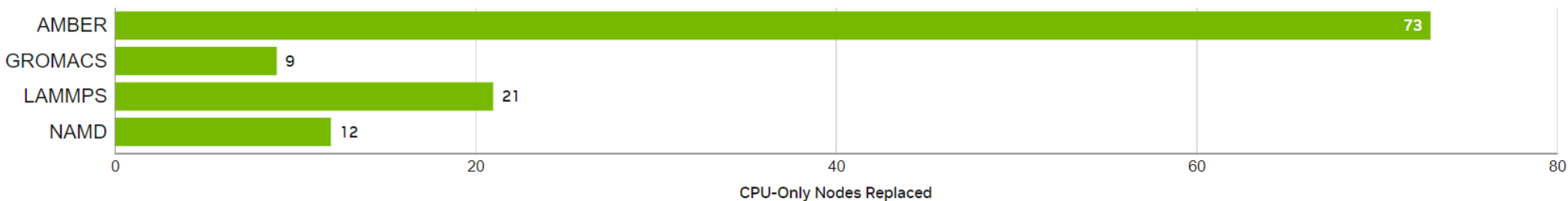
Graphical Processing Units (GPUs) are especially effective for MD

From: developer.nvidia.com/hpc-application-performance

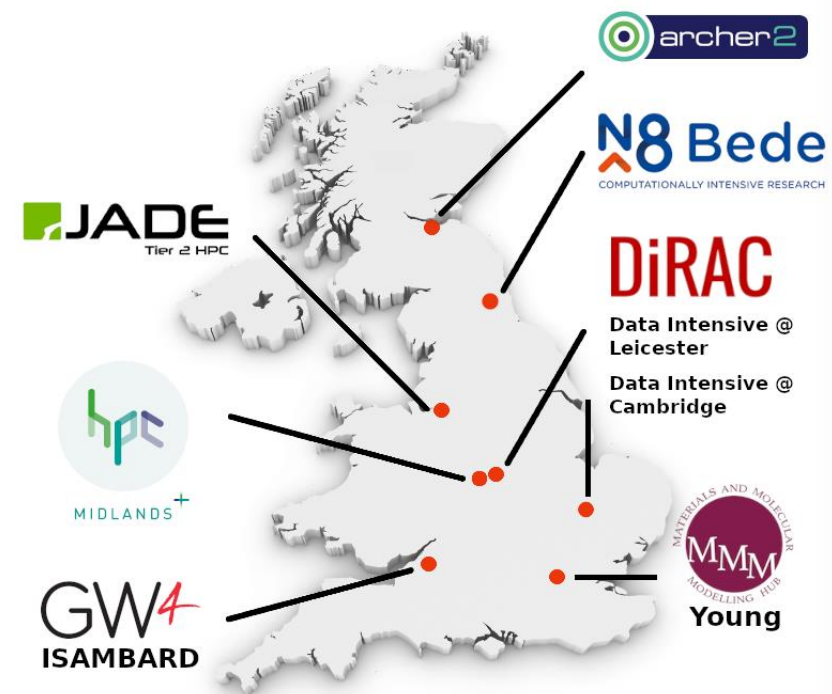
NVIDIA H200



NVIDIA A100



Calculating runtimes: example on UK Tier 2 systems



[as of 2024]

https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator

HECBioSim

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The HECtime Calculator

Enter the following information about your simulation:

JADE2

1000

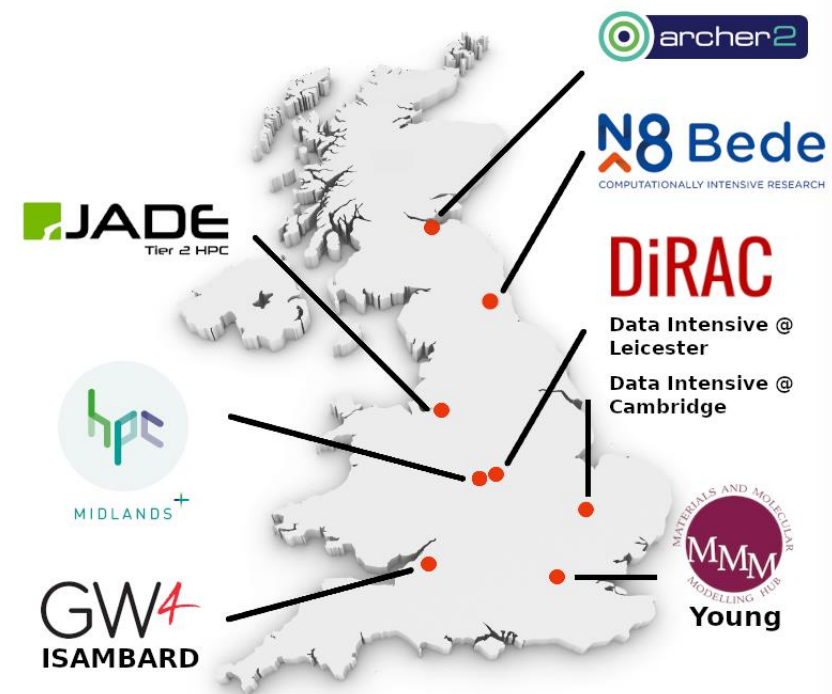
100000

GROMACS 2020.4

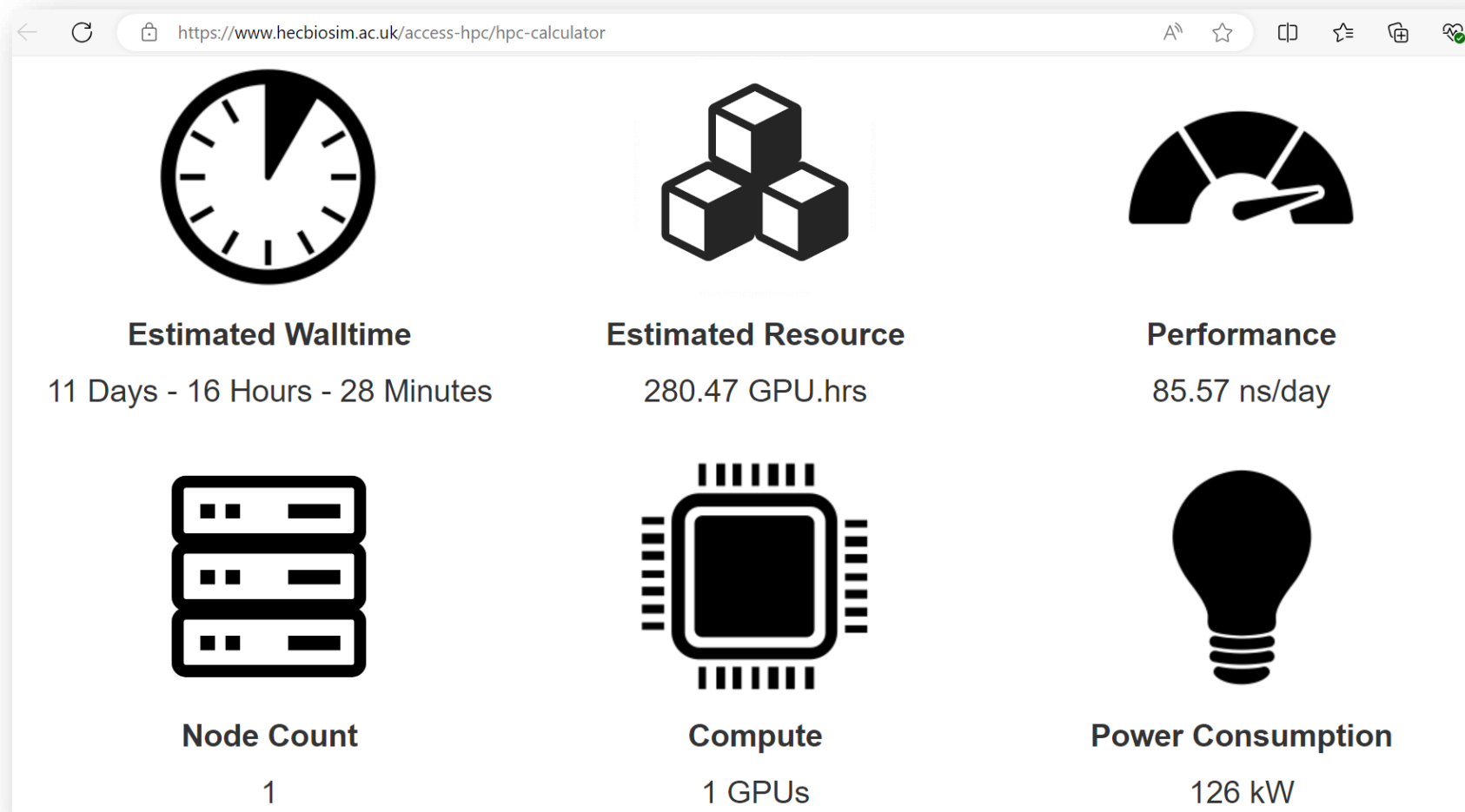
Reset Calculate

<https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator>

Calculating runtimes: example on UK Tier 2 systems



[as of 2024]



<https://www.hecbiosim.ac.uk/access-hpc/hpc-calculator>

Workshop 4: OpenFF Toolkit



OPEN SOFTWARE

Automated infrastructure enables rapid experimentation with minimum human intervention



OPEN DATA

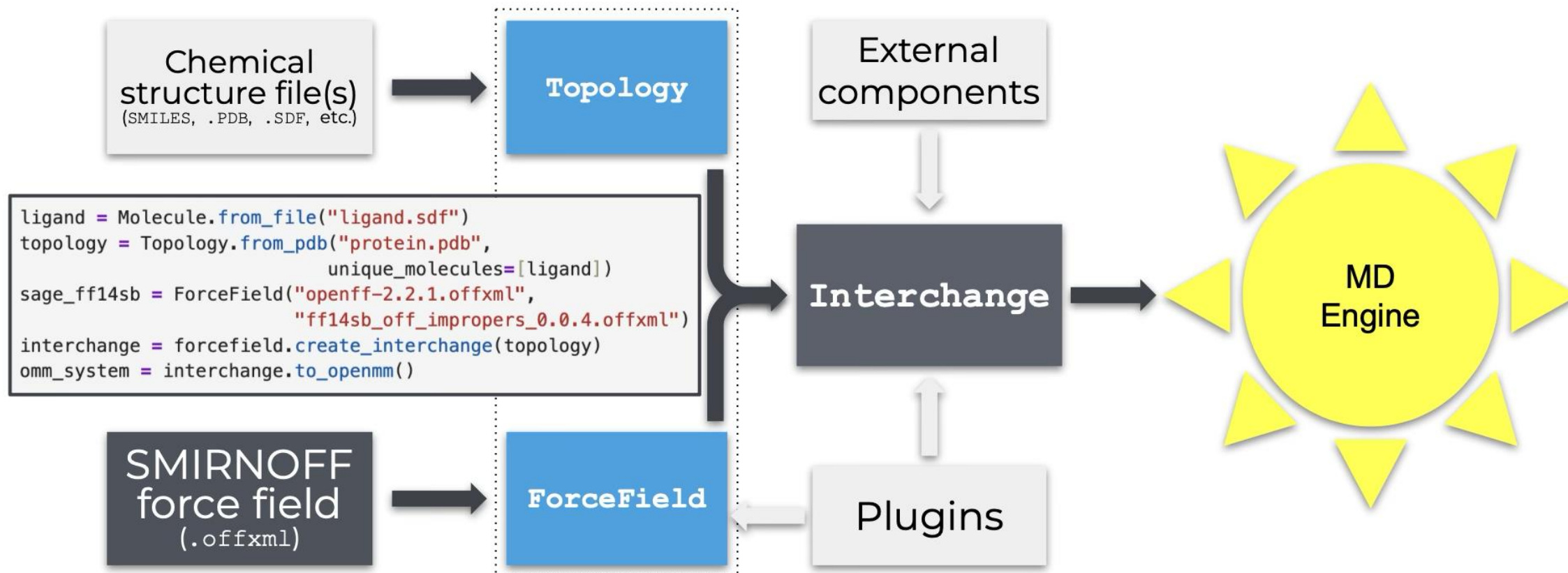
Access to large, high quality experimental and quantum chemical data facilitates easy curation of balanced train / test sets



OPEN SCIENCE

Exploring new force field science:
hypothesis - build
software - train - test -
iterate
is now almost routine

Workshop 4: OpenFF Toolkit



Workshop 4: OpenFF Toolkit



← ↻ 🔒 <https://docs.openforcefield.org/en/latest/examples.html> 🔊 ☆ 📄 ☆ 📌 📌 📌 ...

 **open forcefield**

OpenFF Documentation documentation / Examples

Search 🔍

v: latest

Overview

OpenFF Standards

GETTING STARTED

Installation

Modelling with OpenFF

Examples

PROJECTS

OpenFF Toolkit

Interchange

Units

BespokeFit

QCSubmit

Fragmenter

Tutorials

Tutorials describing key workflows from start to finish, with detailed explanations of what's going on.


open forcefield
Protein-ligand-water systems with Interchange
openff-interchange


open forcefield
Solvate and equilibrate a ligand in a box of water
openff-interchange


open forcefield
Toolkit Showcase: Prepare and run a protein-ligand simulation
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Visualizations in the OpenFF Toolkit
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CONTENTS

Examples

Tutorials

Parametrization and Evaluation

Combining Force Fields

Force Field Development

Uncategorized

Running examples locally

Parametrization and Evaluation

<https://docs.openforcefield.org/en/latest/examples.html>